Worksheet 7.3 — Intersections of Lines & Planes

Grade 12 Calculus and Vectors (MCV4U) · Unit 7: Lines and Planes

The Big Ideas

Lineplane: substitute the line into the plane and solve for t. Compare directions/normals to detect parallel, skew, or coincident.

- Lineplane: substitute x, y, z; solve for t; back-substitute.
- Line $\parallel$ plane when $\vec{d} \cdot \vec{n} = 0$.
- Planes $\parallel$ when normals are parallel.

The Big Picture

What can happen

Two lines, a line and a plane, or two planes can **intersect** in a point or line, be **parallel**, or (lines only) be **skew**.

Solving

Substitute the line's parametric form into the plane's equation, or solve the system of equations, to find any intersection.

Interpreting the outcome

No solution means parallel or skew; a unique solution gives a point; infinitely many means the objects coincide or a line lies in a plane.

Worked Examples

Example 1 — Line meets a plane

Where does $\vec{r} = (1, 0, 0) + t(0, 0, 1)$ meet $z = 5$?

Solution. **Step 1 (write the line's z-component):** $z = 0 + t \cdot 1 = t$.

Step 2 (set it equal to the plane's condition $z = 5$ and solve): $t = 5$.

Step 3 (substitute $t = 5$ into all three components): $x = 1$, $y = 0$, $z = 5$.

Conclusion: they meet at $(1, 0, 5)$.

Example 2 — Substitute into the plane

Where does $x = t, y = t, z = t$ meet $x + y + z = 6$?

Solution. **Step 1 (substitute the line's components into the plane's equation):** $t + t + t = 6$.

Step 2 (solve for t): $3t = 6 \Rightarrow t = 2$.

Step 3 (substitute back): $x = y = z = 2$.

Conclusion: they meet at $(2, 2, 2)$.

Example 3 — Line parallel to a plane

Is $\vec{r} = (0,0,1) + t(1,1,0)$ parallel to $z = 5$?

Solution. **Step 1 (identify the line's direction and the plane's normal):** $\vec{d} = (1,1,0)$; the plane $z = 5$ has normal $(0,0,1)$.

Step 2 (test parallelism, $\vec{d} \cdot \vec{n} = 0$): $(1,1,0) \cdot (0,0,1) = 0 + 0 + 0 = 0$.

Step 3 (rule out the line lying in the plane): the given point $(0,0,1)$ does not satisfy $z = 5$.

Conclusion: parallel (and not contained in the plane).

Example 4 — Parallel lines

Are $\vec{r} = (0,0) + t(2,4)$ and $\vec{r} = (1,1) + s(1,2)$ parallel?

Solution. **Step 1 (compare the direction vectors):** $(2,4)$ and $(1,2)$.

Step 2 (check if one is a scalar multiple of the other): $(2,4) = 2 \cdot (1,2)$.

Conclusion: yes, parallel (the directions are scalar multiples).

Example 5 — Parallel planes

Are $2x + y - z = 1$ and $4x + 2y - 2z = 9$ parallel?

Solution. **Step 1 (read off both normals):** $(2,1,-1)$ and $(4,2,-2)$.

Step 2 (check if one normal is a scalar multiple of the other): $(4,2,-2) = 2 \cdot (2,1,-1)$ — yes, so the planes are parallel.

Step 3 (check if they're actually the same plane by scaling the first equation): $2 \times (2x + y - z = 1)$ gives $4x + 2y - 2z = 2$, which does *not* match the given $= 9$.

Conclusion: parallel and distinct (not the same plane).

Example 6 — Line meets a plane

Where does $x = t, y = 2t, z = 0$ meet $z = 0$?

Solution. **Step 1 (compare the line's z-component to the plane's condition):** the line always has $z = 0$, exactly matching the plane $z = 0$.

Step 2 (interpret): the condition holds for every value of t, not just one.

Conclusion: the line lies entirely in the plane — every point on the line is an intersection point.

Example 7 — Substitute

Where does $x = t, y = t, z = t$ meet $x + y + z = 9$?

Solution. **Step 1 (substitute into the plane's equation):** $t + t + t = 9$.

Step 2 (solve): $3t = 9 \Rightarrow t = 3$.

Conclusion: they meet at $(3,3,3)$.

Example 8 — Line crosses a plane

Is direction $(1,1,1)$ parallel to $x+y+z=4$?

Solution. **Step 1 (identify the plane's normal):** $(1,1,1)$.

Step 2 (test $\vec{d}\cdot\vec{n}=0$): $(1,1,1)\cdot(1,1,1)=1+1+1=3$.

Step 3 (interpret): since the dot product isn't 0, the direction isn't perpendicular to the normal.

Conclusion: no, not parallel — the line crosses the plane.

Example 9 — Simple plane

Where does $\vec{r}=(0,0,0)+t(1,0,0)$ meet $x=4$?

Solution. **Step 1 (write the line's x-component):** $x=0+t=t$.

Step 2 (set equal to 4 and solve): $t=4$.

Step 3 (substitute back): $y=0$, $z=0$.

Conclusion: they meet at $(4,0,0)$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Where does $x = t, y = t, z = t$ meet $x + y + z = 9$?

Question 2

Are the planes $x + y + z = 1$ and $2x + 2y + 2z = 5$ parallel?

Question 3

Is the line with direction $(1, 1, 1)$ parallel to $x + y + z = 4$?

Question 4

Where does $\vec{r} = (0, 0, 0) + t(1, 0, 0)$ meet $x = 4$?

Question 5

Are $\vec{r} = (0, 0) + t(2, 4)$ and $\vec{r} = (1, 1) + s(1, 2)$ parallel?

Question 6

Where does $x = t, y = 0, z = 0$ meet $x = 7$?

Question 7

Are the planes $x - y + z = 0$ and $2x - 2y + 2z = 3$ parallel?

Question 8

Is direction $(1, 0, 0)$ parallel to the plane $y = 5$?

Question 9

Where does $x = 2t, y = t, z = t$ meet $x + y + z = 8$?

Question 10

Are $\vec{r} = (0,0) + t(1,2)$ and $\vec{r} = (0,0) + s(2,1)$ parallel?

Question 11

Where does $\vec{r} = (0,0,1) + t(0,0,1)$ meet $z = 10$?

Question 12

Two planes have normals $(1,2,3)$ and $(2,4,6)$. Parallel?

Question 13 — Challenge

Find where the line $x = 1 + t,\ y = 2 + t,\ z = 3 + t$ meets the plane $x + y + z = 12$.

Question Recap & Answer Key

Question Recap

1. Where does $x = t, y = t, z = t$ meet $x + y + z = 9$?
2. Are the planes $x + y + z = 1$ and $2x + 2y + 2z = 5$ parallel?
3. Is the line with direction $(1, 1, 1)$ parallel to $x+y+z = 4$?
4. Where does $\vec{r} = (0, 0, 0) + t(1, 0, 0)$ meet $x = 4$?
5. Are $\vec{r} = (0, 0)+t(2, 4)$ and $\vec{r} = (1, 1)+s(1, 2)$ parallel?
6. Where does $x = t, y = 0, z = 0$ meet $x = 7$?
7. Are the planes $x - y + z = 0$ and $2x - 2y + 2z = 3$ parallel?
8. Is direction $(1, 0, 0)$ parallel to the plane $y = 5$?
9. Where does $x = 2t, y = t, z = t$ meet $x + y + z = 8$?
10. Are $\vec{r} = (0, 0)+t(1, 2)$ and $\vec{r} = (0, 0)+s(2, 1)$ parallel?
11. Where does $\vec{r} = (0, 0, 1) + t(0, 0, 1)$ meet $z = 10$?
12. Two planes have normals $(1, 2, 3)$ and $(2, 4, 6)$. Parallel?
13. Find where the line $x = 1 + t,\ y = 2 + t,\ z = 3 + t$ meets the plane $x + y + z = 12$.

✓ Answer Key

1. $(3, 3, 3)$
2. parallel and distinct
3. no $(\vec{d} \cdot \vec{n} = 3)$
4. $(4, 0, 0)$
5. yes
6. $(7, 0, 0)$
7. parallel and distinct
8. yes $(\vec{d} \cdot \vec{n} = 0,\ (1, 0, 0))$
9. $4t = 8 \Rightarrow t = 2$: $(4, 2, 2)$
10. no (not scalar multiples)
11. $(0, 0, 10)$
12. yes (normals are multiples)
13. $(1+t)+(2+t)+(3+t) = 12 \Rightarrow 6+3t = 12 \Rightarrow t = 2$: $(3, 4, 5)$